

File permissions in Linux

Project description

Through the Linux commands you can navigate, manage and analyze the file system without the use of graphical user interface (GUI). In this case my organization requires me to update certain files and directory permissions. I used Linux commands to configure authorization or grant the type of access to specific resources in the system. The scenario is to manage the permissions on the files in the `/home/researcher2/projects` directory for the `researcher2` user.

Check file and directory details

There are 3 main commands to check file and directory permissions. `ls -l`: Displays permissions to files and directories. Also displays other additional information, including owner name, group, file size, and the time of last modification. And `ls -la`: Displays permissions to files and directories, including hidden files.

In this case I had to check the permissions on the `projects` directory, To accomplish that first you have to navigate to that directory with the command `cd projects`. After that in this case I used the `ls -la` command to check the file permissions of the directory projects including hidden files, and got the following output:

```
researcher2@85cb09961436:~$ pwd
/home/researcher2
researcher2@85cb09961436:~$ cd projects
researcher2@85cb09961436:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 03:12 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 04:16 ..
-rw--w---- 1 researcher2 research_team  46 Sep  6 03:12 .project_x.t
xt
drwx--x--- 2 researcher2 research_team 4096 Sep  6 03:12 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep  6 03:12 project_k.tx
t
-rw-r----- 1 researcher2 research_team  46 Sep  6 03:12 project_m.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_r.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_t.tx
t
```

Describe the permissions string

In Linux, file permissions are represented with a 10-character string. For a directory with full permissions for the user group, this string would be: `drwxrwxrwx`.

If any character of the 10-character string is represented by a hyphen (-) it means that it does not have that specific permission, the 1st character indicates the file type, a (d) indicates that its a directory, if a hyphen would appear it means its a normal file.

The 2nd, 3rd and 4th characters indicate the permissions for the user. **r** means read permissions, **w** means write permissions and **x** means execute permissions. **rw** is the order for every specific set of permissions. If one of the letters is replaced by a hyphen it means it lacks that specific permission. In this case the user has all 3 permissions. For example if the sequence was **r-x**, it means that the user has only read and execute permissions, but not write permissions.

The 5th, 6th and 7th characters indicate the group permissions, following the same order **rw**.

The 8th, 9th and 10th characters indicate the other permissions, following the same order **rw**.

To explain this better I'll use an example of the output that the bash returned:

```
-rw-rw-r-- 1 researcher2 research_team 46 Sep 6 03:12 project_t.txt
```

The first string of the image are the permissions, after that you can also see the user name, the group name, the date, the time, and finally the file name.

`project_t.txt` has the following permissions:

- The first character is a hyphen, that means it's not a directory but a normal file.
- The user has read and write permissions but not execute permission.
- The group has read and write permissions but not execute permission.
- Other just have read permission.

Change file permissions

The organization does not allow other to have write access to any files. After checking the file and directory permissions we can see that the file with the name `project_k.txt` has the permission for other to have write access. To modify that I used the command **chmod o-w project_k.txt**. The **chmod** command changes permissions on files and directories. Stands for change mode. The **chmod** command requires two arguments. The first argument indicates how to change permissions, and the second argument indicates the file or directory that you want to change permissions for. On the first argument you need to specify which owner type permissions you want to change, **u** for user, **g** for group and **o** for other. Followed by the minus sign (-) if you want to revoke a permission or a plus sign (+) if you want to give that permission, followed by the permission you want to allow or disallow, it can be either **r,w,x** as mentioned earlier. The second argument is the exact name of the file or directory where you want to make permission changes. If there are a lot of changes you can separate them using a comma (,), for example **chmod u-r,g+x project_k.txt** means that I want to revoke the permission to read for the user and add the permission to execute for the group. Next, I'll show the command I used, and how the change has been made.

```
researcher2@85cb09961436:~/projects$ chmod o-w project_k.txt
researcher2@85cb09961436:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep  6 03:12 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_k.tx
t
-rw-r----- 1 researcher2 research_team  46 Sep  6 03:12 project_m.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_r.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_t.tx
t
```

Change file permissions on a hidden file

In this example the research team has archived `.project_x.txt`, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. As we can see on the first screenshot when we checked the file and directory permissions, the user and group both have write permissions and the group doesn't have the permission to read the file. To change that I used the command `chmod u-w,g-w,g+w .project_x.txt`. The next screenshot shows the command and the changes made after reviewing the file and directory permissions, including hidden files with the command `ls -la`.

```
researcher2@85cb09961436:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@85cb09961436:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 03:12 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 04:16 ..
-r--r----- 1 researcher2 research_team  46 Sep  6 03:12 .project_x.t
xt
drwx--x--- 2 researcher2 research_team 4096 Sep  6 03:12 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_k.tx
t
-rw----- 1 researcher2 research_team  46 Sep  6 03:12 project_m.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_r.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_t.tx
t
```

Change directory permissions

In this example the files and directories in the “projects” directory belong to the `researcher2` user. Only `researcher2` should be allowed to access the drafts directory and its contents. Based on the last screenshot, the group has execute permissions in the drafts directory and only the user should be allowed access. To change this, I used the command: `chmod g-x drafts`. By

doing that, the permissions changed and now only the user has access to read, write and execute permissions. The following screenshot shows the command used to change the permissions and also the output received to check if the changes were made, using the `ls -la` command.

```
researcher2@85cb09961436:~/projects$ chmod g-x drafts
researcher2@85cb09961436:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 03:12 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  6 04:16 ..
-r--r----- 1 researcher2 research_team  46 Sep  6 03:12 .project_x.tx
xt
drwx----- 2 researcher2 research_team 4096 Sep  6 03:12 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_k.tx
t
-rw----- 1 researcher2 research_team  46 Sep  6 03:12 project_m.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_r.tx
t
-rw-rw-r-- 1 researcher2 research_team  46 Sep  6 03:12 project_t.tx
t
researcher2@85cb09961436:~/projects$
```

Summary

This document shows how I managed authorization by examining files and directory permissions and then changing permissions on the files and directories for the organization, based on their requirements. Being able to navigate through Linux by using commands is essential in cyber security. In this example I used the commands: `cd`, `ls -l`, `ls -la` and `chmod` to check the file and directory permissions and change them to be as required.